UESTC3031 - Engineering Project Management & Finance (2024 – 2025 Semester 2)

Break-Even and Risk Analysies (Assessment (Report))

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

Purpose of the Assignment

- To demonstrate basics of engineering economics and break-even analysis / make-or- buy analysis

Date Issued to Students: 05th May 2025 / Monday **Deadline / Submission Date: 15th May 2025 / Thursday** Submission: via Moodle

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Read and understood the guidance on plagiarism in the Student Handbook, including the University of Glasgow Statement on Plagiarism	√
Clearly referenced, in both the text and the bibliography or references, **all sources** used in the work	√
Fully referenced (including page numbers) and used inverted commas for **all text quoted** from books, journals, web etc. (Please check with the Department which referencing style is to be used)	√
Provided the sources for all tables, figures, data etc. that are not my own work	√
Not made use of the work of any other student(s) past or present without acknowledgement. This includes any of my own work, that has been previously, or concurrently, submitted for assessment, either at this or any other educational institution, including school (see overleaf at 31.2)	√
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I am aware of and understand the University's policy on plagiarism and I certify that this assignment is my own work**,** except where indicated by referencing, and that I have followed the good academic practices noted above

Signed Chen Yichi /Cao Jiarui .....

Submission Instructions:

- You must submit in **ONE pdf document** (not a zipped file, not two times, not in a word document) the **break-even analysis report** that accompanies it. You already have the instructions in the lecture notes and have been provided the **assignment answering file** that must be used.
- You must prepare this assessment in **pairs/groups of 2**. You must clearly indicate in the file name and in the document the student name's and IDs in the form

Student1_GUID_UESTCID_Student2_GUID_UESTCID.pdf and inside the documents you will list a **table with the student's names, GUIDs, and UESTCIDs of your group members** so that we know who submitted with whom.

- You must submit a **plagiarism declaration** and your submission will go through plagiarism check.
- **No late submissions will be allowed in the Moodle** so if you don't submit you will get a zero for this complete assessment.
- This is a **SUMMATIVE assessment** (i.e. marked assessment).
- Only one student must submit the assessment prepared in groups so we must receive (via Moodle) only 1 submission in the system i.e. one for each group.
- While using references, you must use **IEEE style/format**.

Imran Shafique Ansari

Break-Even and Risk Analyses Report [100 Marks]

Name	GUID	UESTCID
Chen Yichi	[redacted]	[redacted]
Cao Jiarui	[redacted]	[redacted]

Ⅰ. Calculation [45]

You are a production engineer in an electronics company responsible for making electronic circuit boards. A customer has come to you with a design for a microcontroller and has requested you a quote for their business. Your manager has provided you with the documentation from the customer that includes a Bill of Materials (BOM) (as given in the Appendix). He has also provided you with a list of components (see Table I) required for producing the product and their costs from your approved suppliers (as given in the Appendix).

Your manager has requested that you provide a detailed cost analysis for a few different production volumes. *Examine your analysis based on the following:*

1.Pablo is one of the employees in your company, if he borrowed £5000 from a bank at 5% interest, three end-of-year payments will be made.

a. At the end of the first year, he will repay half of the £5000 principal plus the interest that is due.
b. At the end of the second year, he will repay half of the remaining £2,500 principal plus interest due.
c. At the end of the third year, he will repay the final £1,250 principal plus interest due.

Please write the relevant formula and calculate the amount he should repay each year and draw the corresponding Cash Flow Diagram. **[10/100]**

Formula:
Annual Payment = Principal Repayment + Interest (5%)
Let

- P_0=£5 000 be the initial principal,
- r=5%=0.05 be the annual interest rate,
- f=50%=0.5 be the fraction of principal repaid in each of the first two years.

1. General Recurrence for Years 1–2
At the start of year k (for k=1,2), the remaining principal is P_{k-1}
Payment in year kkk consists of:

- Principal repaid: $f P_{k-1}$
- Interest due: $r P_{k-1}$

So the total payment in year k is

$$A_k = f\,P_{k-1} + r\,P_{k-1} = (f + r)\,P_{k-1}, \quad k = 1,2$$

After that payment, the remaining principal at end of year k becomes

$$P_k = P_{k-1} - f\,P_{k-1} = (1 - f)\,P_{k-1}, \quad k = 1,2$$

2. Final (Year 3) Repayment
By the start of year 3, the remaining principal is P_2. In the third year the borrower repays **all** of P_2 plus interest on it, so

$$A_3 = P_2 + r\,P_2 = (1 + r)\,P_2$$

Calculation

- Year 1:

$$A_1 = (f + r)\,P_0 = (0.5 + 0.05) \times 5000 = 0.55 \times 5000 = £2750$$
$$P_1 = (1 - f)\,P_0 = 0.5 \times 5000 = £2500.$$

- Year 2:

$$A_2 = (f + r)\,P_1 = 0.55 \times 2500 = £1375,$$
$$P_2 = (1 - f)\,P_1 = 0.5 \times 2500 = £1250.$$

- Year 3:

$$A_3 = (1 + r)\,P_2 = 1.05 \times 1250 = £1312.50.$$

So the three payments are
A_1=£2 750 A_2=£1375 A_3=£1312.50

Corresponding Cash Flow Diagram



2. When producing a large number of PCBs (usually over 500), it takes a significant amount of time. To meet delivery deadlines, expedited shipping of raw materials is required to produce more than 500 PCBs.

- **Company A**: The expedited shipping cost for raw materials per PCB is £1.5, with no additional charge.
- **Company B**: The expedited shipping cost for raw materials per PCB is £1.3, but an additional £500 container rental fee is required.

Using the component cost figures provided, calculate the total cost to manufacture the following quantities of PCBs: 100, 1000, and 5000. Select the lowest cost method for each production quantity. Your answer must contain sufficient details for your work/calculations to be checked by your manager.

[15/100]

a. Per-board BOM Cost (£/board)

Component	100 (£)	1 000 (£)	5 000 (£)
ATMega328 MCU	1.650	1.090	1.090
CH340G USB–UART	0.343	0.320	0.320
LM1117-3.3 V Regulator	0.605	0.364	0.349
LM1117-5.0 V Regulator	0.605	0.364	0.349
16 MHz Crystal	0.269	0.197	0.192
12 MHz Crystal	0.269	0.197	0.192
LED (Power)	0.052	0.039	0.039
LED (Pin 13)	0.052	0.039	0.039
47 µF Electrolytic ×2	0.298	0.188	0.146
100 nF Ceramic ×4	0.052	0.036	0.028
10 kΩ Resistor ×6	0.042	0.018	0.012
Reset Switch	0.193	0.132	0.124
DC Power Socket	0.877	0.835	0.835
USB Socket	0.324	0.203	0.166
Single-inline Conn. ×2	1.016	0.716	1.244[1]
Dual-in-line Header	0.415	0.303	0.209
Total BOM cost/board	8.078	5.757	5.334

b.Per-board Fabrication & Assembly Costs (£/board)

Quantity	PCB Fab	Assembly
100	0.6904	3.7823
1 000	0.44397	0.78538
5 000	0.36110	0.47481

c. Shipping Cost

- Company A: £1.50/board
- Company B: £1.30/board + £500 one-time fee

d. Total Cost Formulas

For run size Q:

$\text{Total}_A(\boldsymbol{Q}) = \left(\boldsymbol{C}_{\text{BOM}} + \boldsymbol{C}_{\text{fab}} + \boldsymbol{C}_{\text{assy}}\right)\boldsymbol{Q} + \mathbf{1.50}\,\boldsymbol{Q}$,

$\text{Total}_B(\boldsymbol{Q}) = \left(\boldsymbol{C}_{\text{BOM}} + \boldsymbol{C}_{\text{fab}} + \boldsymbol{C}_{\text{assy}}\right)\boldsymbol{Q} + \mathbf{1.30}\,\boldsymbol{Q} + \mathbf{500}$.

e. Numeric Results & Selection

QQ	Σcost/board	Total A (£)	Total B (£)	Cheaper
100	8.078+0.6904+3.7823 = 12.5507	12.5507×100=1255.07	12.5507×100=1255.07	A or B
1 000	5.757+0.44397+0.78538 = 6.98635	6.98635×1000+1500 = 8 486.35	6.98635×1000+1300+500 = 8 786.35	A
5 000	5.334+0.36110+0.47481 = 6.16991	6.16991×5000+7500 = 38 849.55	6.16991×5000+6500+500 = 37 849.55	B

- Q=100: They are the same.
- Q=1 000: Company A is cheapest (no fixed fee dilution).
- Q=5 000: Company B wins, as its lower per-board rate outweighs the £500 fee.

Thus, select Company A for runs up to 1 000 boards, and Company B for 5 000.

3.You have agreed a deal with your component suppliers, PCB manufacturers, and assembly operation on a total variable cost of £7.50 per board for all volumes between 0 and 5,000 boards. If your company has Fixed Costs of £19,500 and a selling price of £15.00, what is the Margin of Safety if you produce 4,000 boards? And what is the company profit when you produce 4000 boards? Draw a graph to illustrate this point.

[12/100]

a. Definitions

- Fixed Cost (FC): £19 500
- Variable Cost per board (VC): £7.50
- Selling Price per board (SP): £15.00
- Production Level (Q): 4 000 boards

b. Break-Even Point (BEP)

The break-even point is the volume at which total revenue equals total cost (no profit, no loss). It is given by:

$$\text{BEP} = \frac{\text{FC}}{\text{SP} - \text{VC}} = \frac{19\,500}{15.00 - 7.50} = 2\,600.$$

(Formula from Lecture 11) .

c. Margin of Safety (MOS)

Margin of safety measures how far actual sales exceed the break-even volume, as a percentage of actual sales:

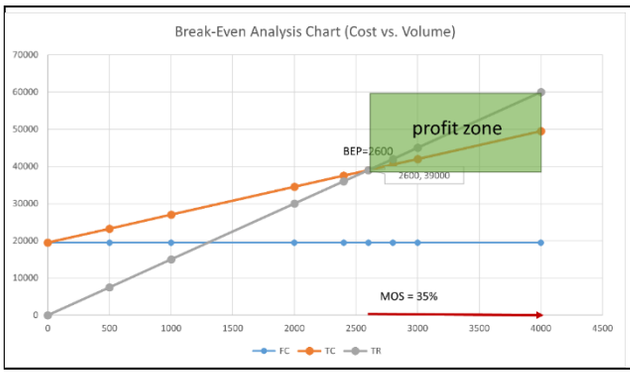
$$\text{MOS} = \frac{Q - \text{BEP}}{Q} \times 100\% = \frac{4\,000 - 2\,600}{4\,000} \times 100\% = 35\%.$$

(Definition from Lecture 11) .

d. Profit at 4 000 Units

Profit can be computed as

$\text{Profit} = (\text{SP} - \text{VC}) \times Q - \text{FC} = (15.00 - 7.50) \times 4\,000 - 19\,500 = 7.50 \times 4\,000 - 19\,500 = 30\,000 - 19\,500 = £10\,500.$



4.Your sales manager advises you that a new customer is willing to pay £27.00 per board but only for a maximum order of 1600 boards. How much profit or loss could you make on this order? Will there be a Margin of Safety in this case? If yes, calculate it. A graph is recommended.
[8/100]

a. Known Data

- **Fixed Cost (FC):** £19 500
- **Variable Cost per board (VC):** £7.50
- **Special Order Price (SP′):** £27.00 per board
- **Order Quantity (Q′):** 1 600 boards

b. Profit Calculation

Use the standard profit formula:

$$\text{Profit} = (\text{SP}' - \text{VC}) \times Q' - \text{FC}$$

Substitute in the numbers:

$$\text{Profit} = (27.00 - 7.50) \times 1600 - 19500 = 19.50 \times 1600 - 19500 = 31200 - 1950 = £11\,700.$$

So the special order yields a profit of **£11 700**.

c. New Break-Even Point (BEP′)

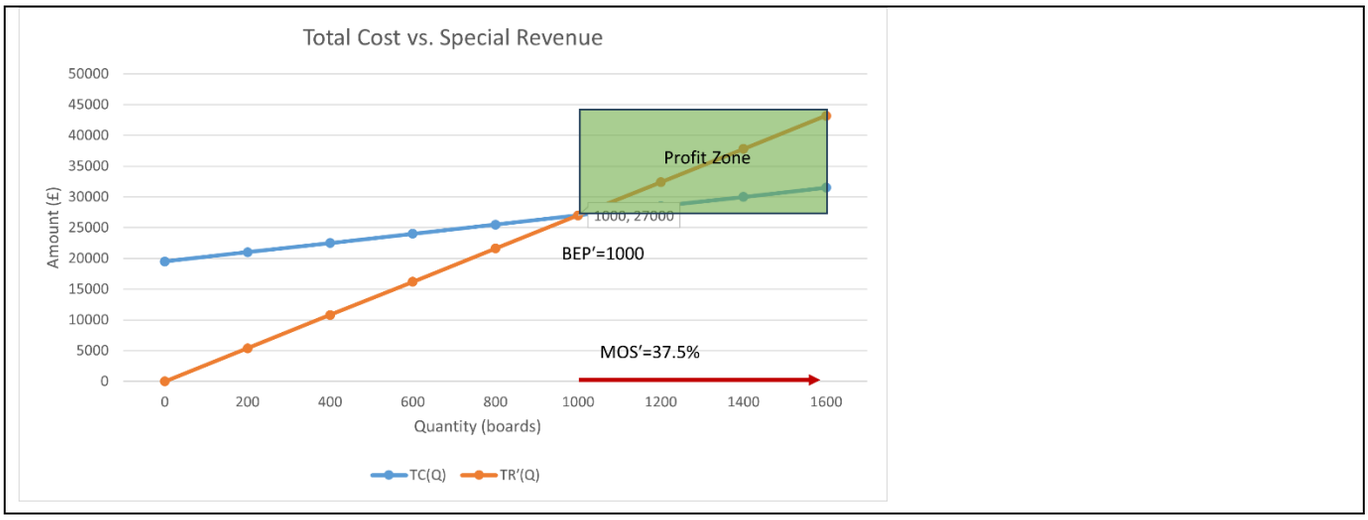
Recompute BEP for the higher price:

$$\text{BEP}' = \frac{\text{FC}}{\text{SP}' - \text{VC}} = \frac{19500}{27.00 - 7.50} = \frac{19500}{19.50} = 1000 \text{ boards}$$

d. Margin of Safety (MOS′)

Since Q′ exceeds BEP′, there is a safety margin:This means that even if sales dropped by 37.5%, the company would still break even.

$$\text{MOS}' = \frac{Q' - \text{BEP}'}{Q'} \times 100\% = \frac{1600 - 1000}{1600} \times 100\% = 37.5\%.$$



Ⅱ. Extensive Discussion [55]

5. Risk Management

Provide 8 possible risks involved in the undertaken project in the form of Risk Register and perform **quantitative analysis** for schedule and cost impacts of listed risks in the form of **influence diagrams** learned in the class, respectively.

[15/100]

A. Risk Register

ID	Risk Statement	Probability	Scope	Quality	Schedule	Cost	Potential Response	Potential Owner
1	Requirements are ambiguous, leading to rework	Medium	High	Medium	Medium	Medium	Host detailed requirements-clarification workshops	Project Manager
2	Key developer becomes unavailable due to illness	Medium	Low	Medium	High	High	Cross-train backup staff and document key modules	HR Manager
3	Selected technology component fails validation testing	Medium	Low	High	High	High	Prototype early and maintain contingency budget	Tech Lead

4	Critical supplier delivers hardware late	High	Low	Low	High	Medium	Qualify and contract a secondary supplier	Procurement Lead
5	Change in regulation requires design modification	Low	Medium	Low	Medium	Medium	Monitor regulatory bulletins; plan design updates	Legal Counsel
6	Project budget cut by finance department	Low	Medium	Low	Medium	Low	Re-prioritize scope; negotiate feature trade-offs	Finance Lead
7	Stakeholder turnover leads to loss of institutional memory	Medium	Low	Medium	Low	Low	Maintain decision logs; conduct hand-over meetings	PMO Coordinator
8	Late discovery of major defects during system integration	Medium	Low	High	High	High	Implement incremental integration testing	QA Manager

B. Quantitative Risk Analysis as Influence Diagrams

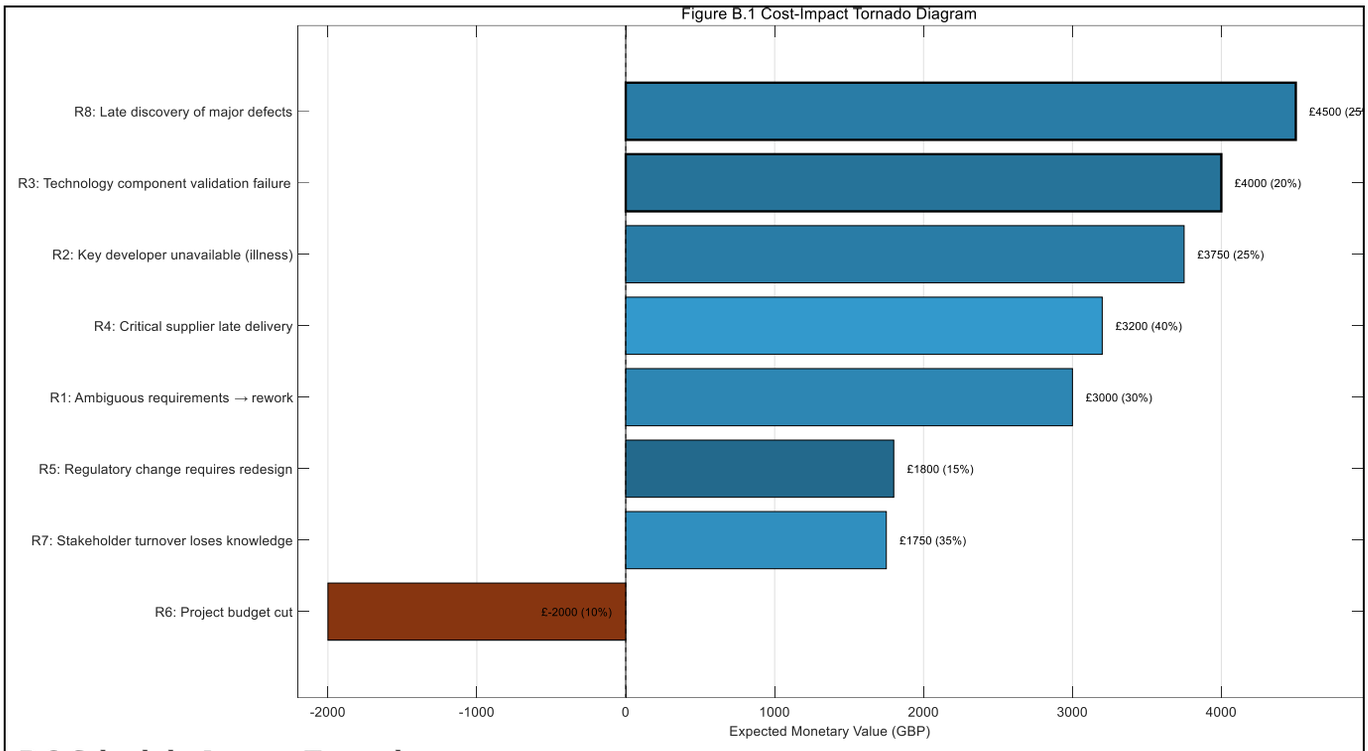
B.1Cost-Impact Tornado

Compute each risk's **EMV** (Expected Monetary Value)

$EMV_i = P_i \times CostImpact_i$

ID	PP	Cost Impact	EMV (
R8	25%	18 000	4 5
R3	20%	20 000	4 0
R2	25%	15 000	3 7
R4	40%	8 000	3 2
R1	30%	10 000	3 0
R5	15%	12 000	1 8
R7	35%	5 000	1 7
R6	10%	-20 000	-2 0

Figure B.1 Cost-Impact Tornado Diagram



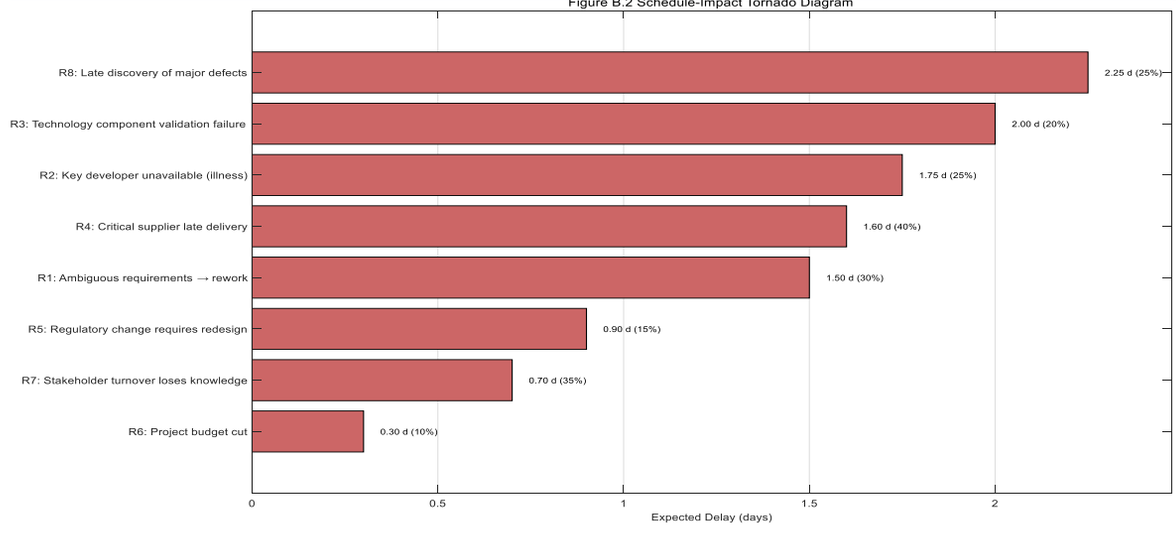
B.2 Schedule-Impact Tornado

Compute each risk's **expected delay**:

$\Delta_i = P_i \times \text{Delay}_i$

ID	PP	Delay (days)	Exp. Delay
R8	25%	9	
R3	20%	10	
R2	25%	7	
R4	40%	4	
R1	30%	5	
R5	15%	6	
R7	35%	2	
R6	10%	3	

Figure B.2 Schedule-Impact Tornado Diagram



The following questions are based on the same background, for each (sub) question, the length must be no longer than 250 words.

6. Suppose you are the owner of Innovation Solution Inc., a start-up that has become highly successful by designing and selling specialized high-performance network routers **(Product Line A)** primarily to large enterprise data centers, the company is now planning to diversify its product range to create new profit points. The new product lines include **Product Line B**: Consumer-grade smart home hubs, which require faster development cycles, different marketing channels, and lower-cost manufacturing; and **Product Line C**: Custom IoT sensor arrays for specific industrial clients, which require close collaboration between specialized engineers, sales acting as consultants, and flexible operations for bespoke builds. How should the organizational structure of the company be adjusted to accommodate these new product lines and projects?

[10/100]

To effectively manage three distinct product lines， A: Enterprise Routers; B: Consumer Smart Home Hubs; C: Customized Industrial IoT Sensors, a hybrid matrix-divisional organizational structure is recommended:

Divisional Structure: Separate divisions for each product line with dedicated marketing, R&D, and operational teams. This can swiftly address specific market needs and product life cycles [1].

Centralized Functions: Core functions, which includes finance, HR, legal, and platform development, are centralized for efficiency, consistency, and resource optimization.

Matrix Management: The matrix management approach aligns with Chandler's principle that structure follows strategy across different growth phases and encourages innovation [2].

This structure balances responsiveness with efficiency, reduces redundancy, and promotes strategic alignment across diverse product lines.

7.For the custom IoT sensor arrays (**Product Line C**), what factors should Innovation Solution Inc. consider when deciding whether to make the sensors in-house or buy them from an external supplier? How would the company evaluate the need for control over the manufacturing process, quality, proprietary information, and the relationship with potential partners?

[10/100]

For Product C, several factors should inform the make-or-buy decision:

a) **Relevant Costs:** Evaluate direct production costs like equipment depreciation, labor and R&D versus external procurement costs, which includes purchase price, logistics, and management overhead [3].
b) **Quality Control and IP Protection:** Internal production enhances quality assurance and secures proprietary technology. However, outsourcing carries IP risk.
c) **Capacity & Lead Times:** Assess internal capacity against peak demands and evaluate the reliability of suppliers' delivery schedules.
d) **Strategic Relationships:** Evaluate suppliers' reliability, financial stability. Ensure they alignment with long-term strategic goals.
e) **PV Analysis:** Discount CapEx and OpEx flows to ensure make-or-buy decisions account for time value of money [4].
f) **Economies of Scale:** Outsourcing benefits large-scale production. Inversely, in-house may suit highly customized or low-volume production.

The company would specifically evaluate control needs by:

Using the make-or-buy framework, Innovation Solution Inc. compares internal manufacturing cost against supplier pricing to determine whether process oversight and quality justify in-house production. A TCO analysis quantifies lifetime capital and operating expenses in present-value terms, which can expose hidden outsourcing costs. For proprietary information, IP sensitivity is assessed and strict NDAs plus data encryption are required to mitigate leakage risk. To ensure strategic alignment, potential partners are scored on financial stability, ISO certification, capacity, and cultural fit by a cross-functional committee.

A comparative cost analysis and sensitivity analysis can further clarify the decision.

8.Innovation Solution Inc. has diversified its product portfolio by adding consumer-grade smart home hubs (**Product Line B**) and custom IoT sensor arrays (**Product Line C**) to an already successful enterprise router business (**Product Line A**). Looking ahead: **[10/100]**

(a) Identify and Analyze New Market Opportunities:
How should the company identify and evaluate emerging market trends, technologies, or customer segments to further expand its business? Consider the role of market research, competitive analysis, and customer feedback in determining the direction for future expansion.

To effectively identify new market opportunities, the company should systematically apply strategic analytical frameworks such as PEST (Political, Economic, Social, and Technological analysis) and Porter's Five Forces model to understand external market dynamics [2]. It should regularly scan the technological advancements such as 5G, IoT, edge computing, and smart infrastructure, which highlights emerging areas of potential competitive advantage. Johnson and Scholes' strategic framework suggests configuring resources to secure long-term advantage through aligned stakeholder engagement. Sun Tzu's philosophy underlines that deep understanding of customer needs and competitors is crucial before market entry. Additionally, to identify customer pain points and preferences, the company must actively engage in market research initiatives and leverage tools such as consumer surveys, focus groups, and data analytics. Utilizing trend forecasting methodologies and scenario planning exercises can further help predict market shifts. This can help the company to proactively position itself for future growth opportunities. Maintaining agile and adaptive product development processes will also facilitate the rapid transition of these identified opportunities into viable commercial offerings [2]. As a result, the firm's ability to capture market share swiftly can be enhanced.

(b) Role of Strategic Partnerships and Acquisitions:
Discuss how forming strategic partnerships, joint ventures, or pursuing mergers and acquisitions could facilitate rapid entry into new markets or product areas. What criteria should be used to select the right partners or targets, and how can these move support both short-term profitability and long-term innovation?

Strategic partnerships, joint ventures, and mergers and acquisitions are critical for quickly accessing new technologies, markets, and distribution channels. Through leveraging existing partner strengths and established market presence, these strategies can significantly facilitate rapid entry into new markets or product areas. Forming strategic partnerships allows the sharing of risks and costs associated with entering unfamiliar markets, while joint ventures provide opportunities to utilize combined expertise, resources, and technologies. Therefore, the development cycles are quickened, and competitive advantages are enhanced. Integrating technology strategy, targets should be chosen for both product and manufacturing fit, ensuring M&A supports core competencies and avoids over-diversification risks. The company should establish a systematic process for identifying and evaluating potential partners or acquisition targets based on several criteria. This includes technological complementarity, market strength, operational compatibility, and cultural alignment. For the company, conducting rigorous due diligence that assesses the financial stability, management quality, operational resilience, innovation capability, and market reputation of potential partners is essential. Additionally, clearly structured partnership agreements with explicit performance metrics and objectives ensure alignment of interests and effective long-term collaboration. These move support short-term

profitability by rapidly expanding market reach, enhancing operational capabilities, and optimizing resource utilization. They can also boost short-term revenue through cross-selling and integrating complementary product portfolios. In terms of long-term innovation, strategic collaborations and acquisitions provide access to advanced technologies, R&D expertise, and novel business practices. This fosters a culture of innovation and strengthens the company's competitive positioning in evolving markets. Ultimately it can sustain future growth and innovation [2].

(c) Balancing Innovation with Scalability:
As Innovation Solution Inc. grows its product mix, what measures should be in place to ensure that increased diversification does not dilute its core competencies and innovation capabilities? How might investments in R&D or technology platforms contribute to a sustainable competitive advantage?

Balancing innovation with scalability necessitates establishing a robust innovation management system that encourages creativity while maintaining operational efficiency. Implementing modular technology architectures and standardized product platforms can allow scalable growth of product mix and facilitate rapid market deployment of innovative solutions at the same time. The company should prioritize investments using comprehensive financial evaluation metrics, including Return on Investment (ROI), Net Present Value (NPV), and Internal Rate of Return (IRR), to ensure projects align with strategic business objectives. Clear governance structures should be set up. It should define innovation processes, set measurable performance objectives, and ensure appropriate resource allocation. Maintaining flexible and agile product development methodologies, such as lean or iterative approaches, allows for rapid experimentation and adjustments without jeopardizing scalability. Regular strategic reviews and structured feedback loops involving cross-functional teams can ensure that innovation initiatives remain closely aligned with corporate goals. Thus, this measure is needed. Investments in R&D or technology platforms directly contribute to sustainable competitive advantage by enabling continuous innovation, differentiation, and improved product performance [2,4]. They help establish unique proprietary capabilities that competitors find difficult to replicate. As a result, the competitive advantage increases. The company should regularly update and expand the company’s technological foundations and investments. This enhances responsiveness to market changes and improve customer satisfaction through innovative solutions that tailor to evolving market demands. Regular strategic reviews and structured feedback loops involving cross-functional teams ensure that these investments remain closely aligned with corporate goals. Consequently, long-term competitive positioning and profitability can be sustained.

9.Given the current operational shifts—serving a mix of high-performance enterprise products, consumer electronics, and custom industrial IoT solutions—Innovation Solution Inc. will face challenges in balancing efficiency, flexibility, and control as it grows. Consider the following: **[10/100]**

(a) Decentralization vs. Centralization:
Analyze the trade-offs between a centralized organizational structure and a decentralized (or matrix) organization for managing diverse product lines. How might each structure impact communication, decision-making, and responsiveness to market dynamics across the different business units?

Companies with diverse product lines must choose between a **centralized** structure and a **decentralized** (or matrix) structure. Each of this has advantages and drawbacks.

Communication

In centralized organizations, information flows top-down through a clear hierarchy. This ensures consistent messaging under a unified vision. Yet over-centralization exemplifies Greiner's second stage direction crisis, heavily hindering agile tactical decisions. A decentralized or matrix approach encourages more direct communication across product teams, which promote knowledge sharing and quicken issue resolution. Nevertheless, the matrix structure breaks down functional silos, so complexity in communication increases.

Decision-Making

Centralized structures reserve major decisions for top management. This guarantees the utilization of company-wide analyses, for example, **make-or-buy** cost studies or **DCF** project appraisals. However, the approach is rigorous and unified, which can slow decision cycles. By contrast, decentralized structures empower managers to act quickly on local market needs. For instance, a business unit could outsource a component to a specialized supplier to meet a deadline without waiting for headquarters. Nevertheless, if units diverge from the overall strategy, inconsistent decisions can be made [1].

Market Responsiveness

Decentralized organizations tend to be more responsive to market dynamics because adjustments happen close to the customer. Each unit can swiftly adapt products to regional trends. Centralized firms can better coordinate large-scale strategic shifts. However, their distance from individual markets means slower response to changes. A matrix structure aims to combine these strengths by sharing resources and information across units, but its complexity can occasionally hamper quick action [2].

(b) Scalable Organizational Design:
What organizational changes (such as establishing product-specific units, cross-functional teams, or dedicated innovation hubs) can support scalable growth as the company continues to expand? Evaluate the benefits and risks associated with each model in terms of maintaining quality, operational control, and proprietary advantages.

Establishing product-specific units can support scalable growth by deepening specialization. Each unit builds in-house expertise and help maintain high quality through focused oversight. Keeping production and development internal also preserves operational control over processes and timelines. Meanwhile, proprietary knowledge staying in-house can protect intellectual property which could be at risk if core work is outsourced. However, this model can create silos or duplicate efforts, so strong coordination is needed to ensure consistent quality and avoid fragmentation [1].

Cross-functional teams, which unite staff from different functions into one project team, are another approach. By incorporating diverse expertise early, these teams improve quality and catch issues quickly within the organization. Keeping development internal in this way also means the company isn't reliant on external contractors, which can introduce some issues. All critical knowledge stays within the firm, preserving proprietary know-how. However, managing cross-functional groups can be complex. competing functional priorities or unclear accountability may hurt efficiency if not well managed [2].

Finally, establishing dedicated innovation hubs can support scalable growth by driving continuous innovation. These hubs let the company cultivate new technologies internally and keep innovation in-house. Thus, the company can control over key developments. It also strengthens competitive position due to new IP stays within the company. An innovation hub does require substantial investment. However, by using discounted cash flow analysis, its future benefits justify the upfront costs. Risks include uncertainty of research outcomes and potential disconnect from the main business if the hub is too isolated [4].

(c) Integration of Engineering and Finance Considerations:

In your answer, highlight how financial planning, budgeting, and performance metrics can be integrated within the organizational structure to align with strategic objectives. How can the company ensure that engineering innovations are efficiently transitioned to scalable production and market success without losing operational efficiency?

Financial planning and engineering design must be interwoven to ensure innovations meet strategic and financial objectives. The Introduction to Engineering Economics class emphasizes using an economic decision framework to ensure technical projects support business objectives. This means engineers and finance teams collaborate from the start. Applying CapEx/Opex and TCO concepts, all projects should pass PV-based stage-gate reviews before scaling to production, linking finance and engineering. Organizationally, the company can embed financial analysts within engineering project teams and implement stage-gate reviews with financial checkpoints. The innovations progress can be made only if it meets budget and ROI criteria. The Discounted Cash Flow class demonstrates financial tools like Net Present Value (NPV) analysis to help quantify an innovation's expected future returns in today's dollars. If an initiative's discounted future cash flows exceed its costs, it aligns with strategic value creation goals. Internal Rate of Return (IRR) and Return on Investment (ROI) metrics further gauge a project's profitability and can guide portfolio choices. Meanwhile, the Make or Buy decisions illustrate cross-functional evaluation of developing in-house vs. outsourcing. Such decisions hinge on cost-benefit comparisons and capacity considerations, which ensures new products can be scaled without straining operations. Throughout development, performance metrics track both innovation impact and operational efficiency. ROI and IRR indicate financial returns, while break-even analysis shows how many sales are required to cover costs before profits begin [5]. By integrating these financial metrics, the company can ensure that projects are economically viable, and innovations are smoothly transitioned into efficient production and market success.

References

[1] “Lecture 9 – Introduction to Engineering Economics,” Engineering Economics course, PowerPoint slides, May 2025.
[2] “Lecture 13 – Engineering Economics (TCO Capex Opex),” Engineering Economics course, PowerPoint slides, May 2025.
[3] “Lecture 13 – Engineering Economics (Discounted Cash Flow),” Engineering Economics course, PowerPoint slides, May 2025.
[4] “Lecture 10 – Engineering Economics (Make or Buy),” Engineering Economics course, PowerPoint slides, May 2025.
[5] “Lecture 11 – Engineering Economics (Break-Even Analysis),” Engineering Economics course, PowerPoint slides, May 2025.

APPENDIX: Arduino UNO R3 BOM Costs

BOM Prices for 2025 (All prices in UKP)			
LM1117 (5.0V or 3.3V)			
Quantity			Price (each)
1+			0.916
50+			0.795
100+			0.605
250+			0.565
500+			0.525
1000+			0.364
2500+			0.356
5000+			0.349
Crystal (16MHz or 12MHz)			
Quantity			Price (Each)
1+			0.444
50+			0.327
100 +			0.269
250 +			0.260
500 +			0.251
1000 +			0.197
5000 +			0.192
ATMega 328 Microcontroller			
Quantity			Price(each)
1+			2.35
10+			2.06
25+			1.85
50+			1.72
100+			1.65

250+			1.49
500+			1.30
1000+			1.09

CH340G USB – Serial Interface			
Quantity			Price(each)
1+			0.507
10+			0.391
30+			0.367
100+			0.343
500+			0.332
1000+			0.320

USB Socket			
Quantity			Price(each)
5+			0.464
75+			0.324
150+			0.273
250+			0.216
500+			0.203
1500+			0.190
2500+			0.178
5000+			0.166

DC Power Socket			
Quantity			Price(each)
1+			1.441
10+			1.069
25+			0.983
100+			0.877
250+			0.856
500+			0.835

47uF / 25V Electrolytic SMD			
Quantity			Price(each)
10+			0.219
100+			0.149
500+			0.094
2500+			0.086
5000+			0.073
10000+			0.070
50000+			0.063

100nF / 10V ceramic 0603 SMD			
Quantity			Price(each)

10+			0.024
100+			0.013
500+			0.011
1000+			0.009
2000+			0.008
4000+			0.007
20000+			0.006

10K Resistor / 0603 SMD.			
Quantity			Price(each)
10+			0.010
100+			0.007
500+			0.004
1000+			0.003
2500+			0.002

LED SMD RED			
Quantity			Price(each)
5+			0.102
25+			0.054
100+			0.052
250+			0.045
500+			0.040
1000+			0.039

Single In Line connector (8 way) [Assume same cost for 6 way and 10way]			
Quantity			Price(each)
1+			0.722
10+			0.669
100+			0.508
500+			0.437
1000+			0.358
5000+			0.311
10000+			0.307

6-Way, Dual In-Line header, Through Hole			
Quantity			Price (each)
1+			0.480
10+			0.450
50+			0.415
500+			0.303
1500+			0.260
2500+			0.209

SMD Switch (Reset)			
Quantity			Price
10+			0.193
500+			0.166
1000+			0.132
5000+			0.124

10000+			0.119

PCB Cost: 2 layer, 53 x 68mm. (SEEED)			
Qty			Price(UKP)
5			6.15
10			9.40
50			44.38
100			69.04
500			247.70
1000			443.97
2500			990.05
5000			1805.48
Assembly costs: 28 components; 7 thru' hole, 18 SMT (PCBWAY)			
Qty			Price (UKP) (for the given quantity)
10			266.40
50			312.40
100			378.23
500			625.32
1000			785.38
2500			1600.02
5000			2374.05

Table I: List of Components, and their Quantities

Component	Description	Quantity
ATMega328	Microcontroller IC	1
CH340G	USB to Serial IC	1
LM1117-3.3	Linear Regulator IC (3.3V)	1
LM1117-5.0	Linear Regulator IC (5.0V)	1
Crystal, 12MHz- HC49 Package	USB Crystal	1
Crystal, 16MHz-HC49 Package	CPU Crystal	1
LED1 Red, SMT	LED, Power indication	1
LED2, Red, SMT	LED Indicator, Pin 13	1
C1, C2	Capacitor, electrolytic, 47uF, 25V	2
C3, C4, C5, C6	Capacitor, decoupling, 100nF, 10V, 0603	4
R1, R2, R3, R4, R7, R8	Resistor, 10K, 0603	6
SWT_1	Switch, NO, reset, SMT	1
CONN_1	DC Power Connector, Through hole	1
CONN_2	USB Connector, Through hole	1
SIP_1	8-Way Single In-Line Connector, Through Hole	2
SIP_2	6-Way, Single In-Line Connector, Through Hole	1
SIP_3	10-Way, Single In-Line Connector, Through Hole	1
SIP_4	6-Way, Dual In-Line header, Through Hole	1
PCB_1	Printed Circuit Board, FR-4, Double sided, 1.6mm (53mm x 68mm)	1
ASSY_1	Board Assembly costs; 28 Component, 7 Through Hole	1

----------END OF ASSESSMENT ----------